Direct On Line 2 Speed Motor MAPS Template (Electrical)

The Direct On Line 2 Speed Motor MAPS template is designed to control a motor that is connected as a DOL, which is controlled by two outputs: one for slow speed and another for high speed.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Advanced Direct On Line 2 Speed Motor MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced DOL 2 Speed Motor Functional Philosophy

The DOL start slow and fast speed outputs are enabled once all interlocks are ready and a start request is received.

Field Interlocks

- MCC (Overload) Trip Feedback
- Isolator On Feedback
- Safety Interlock Feedback
- Motor Current Feedback (Analog Input) (Current above High Limit Setpoint)

PLC Interlocks

- Start Interlock Ready
- Process Interlock Ready
- Restart Delay Interlock
- There are no interlock faults latched

Motor Current – The motor current is monitored constantly after the running setpoint time has elapsed, if the current goes above the high current limit for five seconds the motor will trip on overcurrent.

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

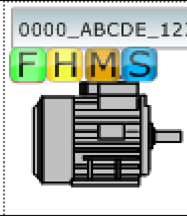
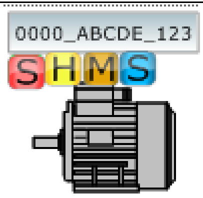
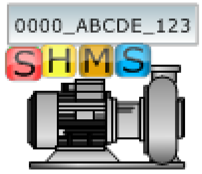
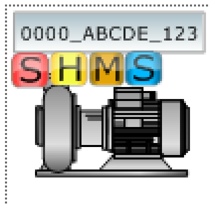
Maintenance mode - will disable the Start and Process Interlocks, but respects all the other interlocks. This is used to test a DOL while the process is stopped.

Simulation mode - will disable the physical output of the DOL and simulate an input run feedback after the elapsed setpoint time from the faceplate. The Start and Process Interlocks are respected. Simulation mode is used to test the process logic without starting the physical device itself. It is very useful in testing, commissioning and maintenance.

Start and stop requests are initiated by selecting a mode on the DOL faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode & Desk mode** - the operator can start and stop from the faceplate.
- **Manual mode & Field mode** - the motor can be started and stopped from the start and stop push buttons in the field.

Typical Faceplate Graphics for the Advanced DOL 2 Speed Motor MAPS Template

DOL ISO	DOL MOTOR LEFT	DOL MOTOR RIGHT	DOL PUMP LEFT	DOL PUMP RIGHT
				

F	Fast Speed Indication
S	Slow Speed Indication
H	Manual Mode
M	Maintenance Mode
S	Simulation Mode
0000_ABCDE_123	Motor Control Name

Graphics legend:

Line colour	Fill colour	DOL status
Black	Grey	Stopped
Grey	White	Stopping
Green	White	Starting
Black	Green	Running
Black	Yellow/Red	Fault Active

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	**Marshal**	**x**		
SSCL 1	Fail To Start Setpoint	**Analog**	**x**		
SSCL 2	Fail To Stop Setpoint	**Analog**	**x**		
SSCL 3	Running Setpoint	**Analog**	**x**		
SSCL 4	Stopping Setpoint	**Analog**	**x**		
SSCL 5	Simulate Setpoint	**Analog**	**x**		
SSCL 6	Restart Setpoint	**Analog**	**x**		
SSCL 7	Max Current Setpoint (High Limit)	**Analog**	**x**		
SSCL 8	Min In Current Setpoint (Raw Min)	**Analog**	**x**		
SSCL 9	Max In Current Setpoint (Raw Max)	**Analog**	**x**		
SSCL 10	Min Out Current Setpoint (Eng. Min)	**Analog**	**x**		
SSCL 11	Max Out Current Setpoint (Eng. Max)	**Analog**	**x**		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	**Marshal**	**x**		
SSSL 1	Starting Time Process Value (actual)	**Analog**	**x**		
SSSL 2	Stopping Time Process Value (actual)	**Analog**	**x**		
SSSL 3	No of Operations (actual)	**Analog**	**x**		
SSSL 4	Restart Delay Time Remaining (actual)	**Analog**	**x**		
SSSL 5	Motor Current Process Value (actual)	**Analog**	**x**		
SSSL 6	Total Running Hours	**Analog**	**x**		

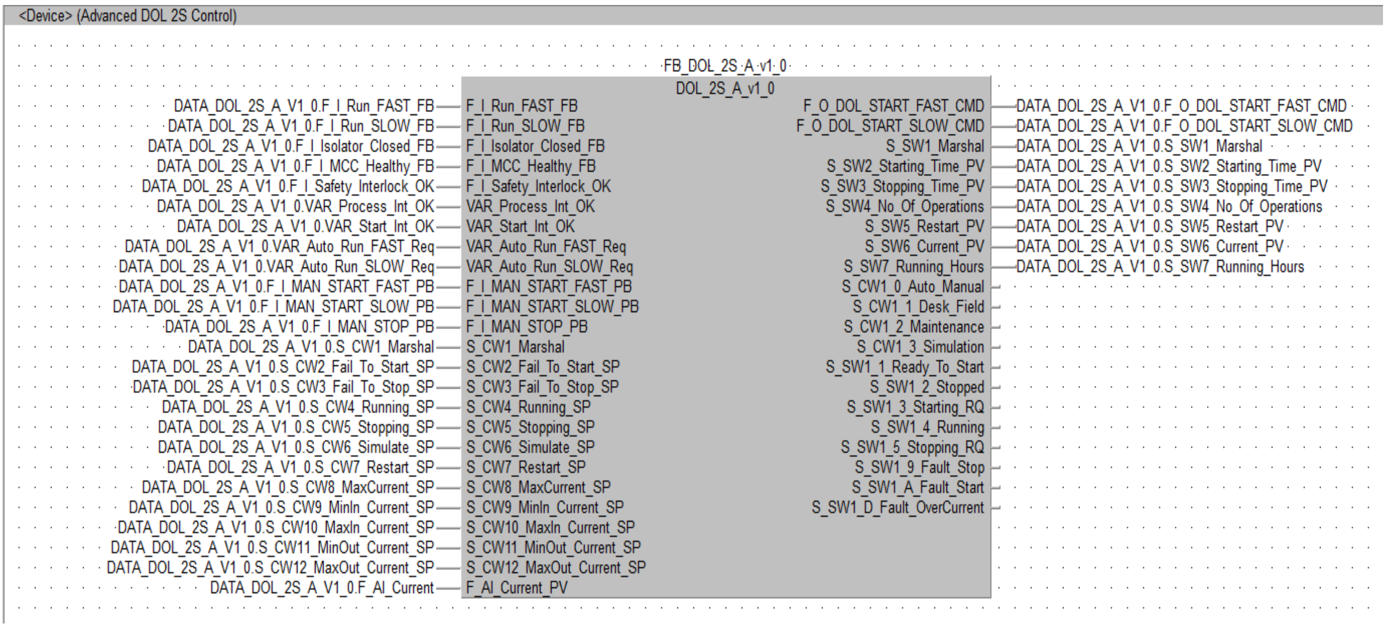
Digital Inputs:		Advanced	Standard	Basic
DI 0	Running Signal Fast	**x**		
DI 1	Running Signal Slow	**x**		
DI 2	Isolator Healthy (On = Healthy)	**x**		
DI 3	MCC Healthy (On = Healthy)	**x**		
DI 4	Safety Interlock Healthy (On = Healthy)	**x**		
DI 5	Manual Start Push Button Fast	**x**		
DI 6	Manual Start Push Button Slow	**x**		
DI 7	Manual Stop Push Button (Off = Stop)	**x**		
Digital Outputs:				
DO 0	Device Start Fast	**x**		
DO 1	Device Start Slow	**x**		
Analog Input:				
AI 0	Analog Input for Current Drawn	**x**		

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Auto/Manual	**x**	**x**		
Bit 1	Field/Desk	**x**	**x**		
Bit 2	Maintenance	**x**	**x**		
Bit 3	Simulation	**x**	**x**		
Bit 4	Alarms All Reset		**x**		
Bit 5	Reset No Operations Accumulator	**x**	**x**		
Bit 6	Start Command	**x**	**x**		
Bit 7	Reset Running Hours Accumulator	**x**	**x**		
Bit 8	Stop Command	**x**	**x**		
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Fast Speed Command	**x**	**x**		
Bit 15	Slow Speed Command	**x**	**x**		

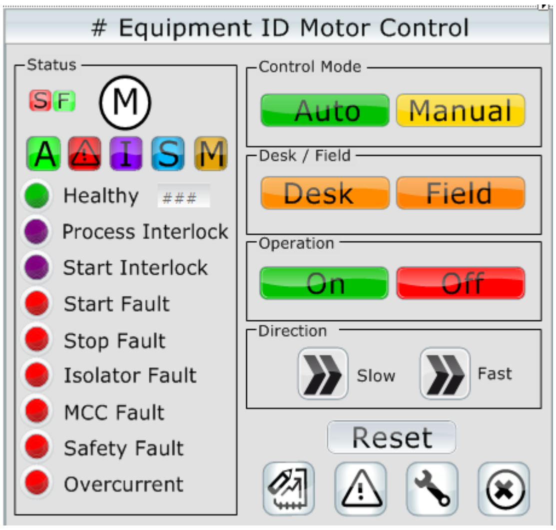
SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Process Interlock		x		
Bit 1	Ready To Start		x		
Bit 2	Stopped		x		
Bit 3	Starting		x		
Bit 4	Running		x		
Bit 5	Stopping		x		
Bit 6	Fault		x		
Bit 7	Start Interlock OK		x		
Bit 8	Isolator Closed Fault	x	x		
Bit 9	Fault Fail To Stop	x	x		
Bit 10	Fault Fail To Start	x	x		
Bit 11	MCC Fault	x	x		
Bit 12	Safety Fault	x	x		
Bit 13	Overcurrent Fault	x	x		
Bit 14	Spare				
Bit 15	Fast / Slow Speed Indication		x		

Resources	Advanced	Standard	Basic
Digital Inputs	8		
Digital Outputs	2		
Analog Inputs	1		
Analog Outputs	0		
PLC Memory Steps	565		
System Words Used	54		
System Bits Used	106		
SCADA SCAN Points	19		

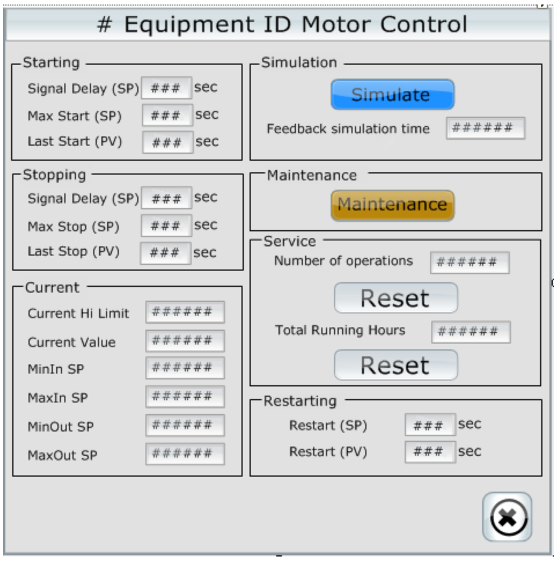
Template: Advanced DOL 2 Speed PLC Function Block

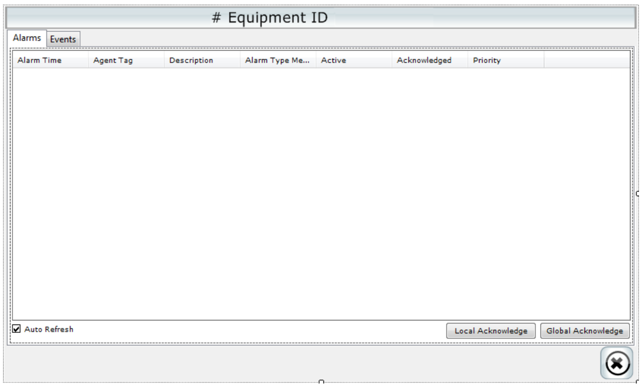


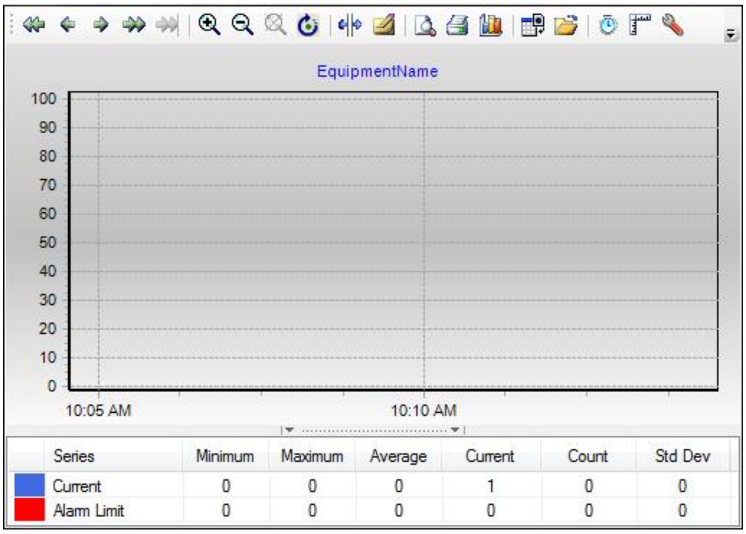
Template Graphic: Advanced Home Screen



Control	Descriptions
Auto	Auto control by PLC
Manual	Manual control by operator, desk and field mode
Desk	Manual control from the SCADA faceplate
Field	Manual control from the field start/stop station
On	Manual start the DOL
Off	Manual stop the DOL
Reset	Reset the DOL after a fault has been repaired
Healthy ###	Healthy to start, ### - Restart delay countdown in seconds
Process Interlock	DOL process ok to run interlock ready
Start Interlock	DOL ok to start interlock ready
Start Fault	DOL failed to start fault
Stop Fault	DOL failed to stop fault
Isolator Fault	DOL isolator off fault
MCC Fault	DOL MCC tripped fault
Safety Fault	DOL Safety off fault
Overcurrent	DOL overcurrent fault
Slow	Slow speed selection and indication
Fast	Fast speed selection and indication

Template Graphic: Advanced Setup Screen	Control	Descriptions
	Simulate	Enable simulation of DOL, disables the physical output
	Feedback Time	Time for simulation to simulate run feedback signal
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of operations	Total number of DOL start operations
	Running Hours	Total number of DOL running hours
	Reset	Reset the total counter of the current service
	Start Signal Delay (SP)	Delay the PLC to accept the DOL is running
	Max Start (SP)	Time for running signal is on after start signal is on
	Last Start (PV)	Last time for running signal was on after start signal was on
	Stop Signal Delay (SP)	Delay the PLC to accept the DOL is stopped
	Max Stop (SP)	Time running signal is off after start signal is off
	Last Stop (PV)	Last time running signal was off after start signal was off
	Current Hi Limit	Sets the value of the maximum current for the DOL to trip at
	Current Value	Actual current the DOL is running at
	MinIn, MaxIn	Raw current input values for scaling
	MinOut, MaxOut	Engineering current values for scaling
	Restart (SP)	Restart prevention time of the DOL setpoint
	Restart (PV)	Remaining time before a DOL restart is allowed

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Current	Display the current of the DOL in the trend, with minimum, maximum and average values.
	Alarm Limit	Display the current Setpoint of alarming on high current of the DOL in the trend, with minimum, maximum and average values.